Summary in Graph

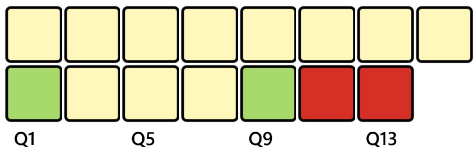
Exam Summary (GO Classes DA Test Series 2026 | Linear Algebra | Topic Wise Test 3)

Qs. Attempted:	**9** (6 + 3)	**Correct Marks:**	**2** (2 + 0)
Correct Attempts:	**2** (2 + 0)	**Penalty Marks:**	**1.33** (0 + 1.33)
Incorrect Attempts:	**7** (4 + 3)	**Resultant Marks:**	**0.67** (2 + -1.33)

Total Questions:	**15** (10 + 5)
Total Marks:	**20** (10 + 10)
Exam Duration:	**45 Minutes**
Time Taken:	**45 Minutes**

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (58) out of 575 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Let V be the subspace of $\mathbb{R}^4$ defined by the equation $x_1 + x_2 + x_3 + x_4 = 0$.
Which of the following is/are TRUE about V ?

A. V is null space of rank-1 matrix
B. V is null space of rank-3 matrix
C. Dimension of V is 3
D. Dimension of V is 1

Your Answer: A;D | Correct Answer: A;C | Incorrect | Time taken: 02min 16sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A and B be matrices of the same size, say $m \times n$. Consider the matrix C formed by stacking A on top of B, i.e.,

$$C = \begin{bmatrix} A \\ B \end{bmatrix}$$

Here, C is a $(2m) \times n$ matrix.
Which of the following is TRUE about the Null space of C?

A. $\text{null}(C) = \text{null}(A) \cap \text{null}\ (B)$
B. $\text{null}(C) = \text{null}(A) + \text{null}(B)$
C. $\text{null}\ (C) = \text{null}(A + B)$
D. $\text{null}(C) = \text{null}(A) \cup \text{null(B)}$

Your Answer: A | Correct Answer: A | Correct | Time taken: 06min 18sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear transformation

$$T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \end{bmatrix}$$

Find $[T]_{\alpha}^{\alpha}$ where α is the standard basis of $\mathbb{R}^2$.

A. $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
B. $\begin{bmatrix} -1 & -2 \\ -1 & -1 \end{bmatrix}$
C. $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$
D. $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 51sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let v_1, v_2, v_3 form a basis α for a vector space V and let $T : V \to V$ be a linear transformation

$$T(v_1) = v_1 - v_2$$
$$T(v_2) = v_2 - v_3$$
$$T(v_3) = v_3 - v_1$$

Find $[T(v)]_{\alpha}$ where $v = v_1 - 2v_2 + 3v_3$.

A. $\begin{bmatrix} 0 \\ 2 \\ -4 \end{bmatrix}$

B. $\begin{bmatrix} -2 \\ -3 \\ 5 \end{bmatrix}$

C. $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

D. $\begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following properties of a matrix are not changed by row operations?

A. The left null space
B. The column space
C. The null space
D. The row space

Your Answer: B;C | Correct Answer: C;D | Incorrect | Time taken: 08min 09sec | Discuss

Q #6 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which of the following transformations of $\mathbf{R}^2$ are linear?

A. $T(\mathbf{v}) = (v_1, v_2)$
B. $T(\mathbf{v}) = (|v_1|, 0)$
C. $T(\mathbf{v}) = (3v_1, v_2 - 3v_1)$
D. $T(\mathbf{v}) = (0, 0)$

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Suppose the matrix $A = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & p & q \end{bmatrix}$ has row echelon form $R = \begin{bmatrix} 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.
Which of the following is/are ALWAYS the basis for the column space of A ?

A. The first and second columns of R.
B. The first and fourth columns of A
C. The first and fourth columns of R.
D. The third and fourth columns of A.

Your Answer: A;B;C | Correct Answer: B;D | Incorrect | Time taken: 01min 50sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Consider a vector $a \in \mathbb{R}^m$. The column space of the outer product ab^T is:

A. $\{x \in \mathbb{R}^m \mid x = ka, k \in \mathbb{R}\}$
B. $\{x \in \mathbb{R}^m \mid x = kb, k \in \mathbb{R}\}$
C. $\{x \in \mathbb{R}^m \mid x = ka + kb, k \in \mathbb{R}\}$
D. $\{x \in \mathbb{R}^m \mid x = ka + lb, k, l \in \mathbb{R}\}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Linear Algebra

Which among the following subsets of $\mathbb{R}^3$ is a subspace?

A. $\left\{ \begin{bmatrix} t \\ s \\ \sin t \end{bmatrix} \middle| \, t, s \in \mathbb{R} \right\}$
B. $\left\{ \begin{bmatrix} t \\ 2t \\ 1 \end{bmatrix} \middle| \, t \in \mathbb{R} \right\}$
C. $\left\{ \begin{bmatrix} t \\ s \\ t+s \end{bmatrix} \middle| \, t \in \mathbb{R}, s \geq 0 \right\}$
D. $\left\{ \begin{bmatrix} t \\ t+s \\ s \end{bmatrix} \middle| \, t, s \in \mathbb{R} \right\}$

Your Answer: B;C;D | Correct Answer: D | Incorrect | Time taken: 00min 48sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let V be a vector space, and suppose that $v_1, v_2 \in V$ and $w_1, w_2 \in V$ are two ordered bases for V. Assume further that

$$w_1 = 6v_1 - 2v_2$$
$$w_2 = 9v_1 - 4v_2$$

Find the change of coordinates matrix from the ordered basis w_1, w_2 to the ordered basis v_1, v_2.

A. $\begin{pmatrix} 6 & 9 \\ -2 & -4 \end{pmatrix}$
B. $\begin{pmatrix} 9 & 6 \\ -4 & -2 \end{pmatrix}$
c. $\begin{pmatrix} -2 & 6 \\ 9 & -4 \end{pmatrix}$
D. $\begin{pmatrix} 6 & -2 \\ 9 & -4 \end{pmatrix}$

Your Answer: A | Correct Answer: A | Correct | Time taken: 03min 05sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let P_{ij} denote the change of basis matrix from Basis B_j to B_i, and $[\mathrm{T}_A]_{B_k}$ represent the linear transformation matrix with respect to basis B_k.
Which of the following statements accurately reflects the relationship between change of basis matrices and linear transformation matrices?

A. $[\mathrm{T}_A]_{B_2} = P_{12}^{-1}[\mathrm{T}_A]_{B_1} P_{12}$
B. $[\mathrm{T}_A]_{B_3} = P_{23}^{-1}[\mathrm{T}_A]_{B_2} P_{23}$

C. $[\mathrm{T}_A]_{B_3} = P_{13}^{-1}[\mathrm{T}_A]_{B_1}P_{13}$
D. $[\mathrm{T}_A]_{B_1} = P_{12}[\mathrm{T}_A]_{B_2}P_{12}^{-1}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation and $T(\mathbf{x}) = A\mathbf{x}$, where A is the standard matrix for T. Which of the following statements must be TRUE?
(i) A is an $n \times m$ matrix.
(ii) If the columns of A are linearly independent, then T is one-to-one.
(iii) If $m > n$, then T is onto.
(iv) If $n > m$, then T is one-to-one.
(v) If the columns of A span $\mathbb{R}^m$, then T is onto.

A. (i), (ii), (iii), (iv), and (v)
B. (i), (ii), and (iii) only
C. (ii) and (v) only
D. (ii), (iii), and (iv) only

Your Answer: D | Correct Answer: C | Incorrect | Time taken: 04min 03sec | Discuss

Q #13 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Which of the following statements is/are true about a matrix A for which the vector $\begin{bmatrix}1\\1\\1\end{bmatrix}$ is a basis for both the left null space and the null space of A ?

A. The sum of the rows of A is a zero vector.
B. The sum of the columns of A is a zero vector.
C. Such matrix is not possible to exist.
D. Possibly the matrix A could be $A = \begin{bmatrix}-1 & -1 & 2\\ 1 & 0 & -1\\ 0 & 1 & -1\end{bmatrix}$.

Your Answer: B;D | Correct Answer: A;B;D | Incorrect | Time taken: 06min 12sec | Discuss

Q #14 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

If, for a 6×9 matrix A, every solution of the homogenous equation $A\mathbf{x} = \mathbf{0}$ can be expressed as $r\mathbf{v_1} + s\mathbf{v}_2 + t\mathbf{v}_3$ for linearly independent $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^9$, then what can one say about existence of solutions to $\mathbf{Ax} = \mathbf{b}$ for any $\mathbf{b} \in \mathbb{R}^6$?

A. For every $\mathbf{b} \in \mathbb{R}^6$, there is an $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$.
B. For no $\mathbf{b} \in \mathbb{R}^6$, there is an $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$.
C. There may exist some $\mathbf{b} \in \mathbb{R}^6$ for which there is an $\mathbf{x} \in \mathbb{R}^9$ such that $A\mathbf{x} = \mathbf{b}$, but not for every $\mathbf{b}$.
D. The existence of solutions for $A\mathbf{x} = \mathbf{b}$ depends on whether $\mathbf{b}$ is in the null space of A.

Your Answer: D | Correct Answer: A | Incorrect | Time taken: 10min 33sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Linear Algebra

$$A = \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 4 & 2 & 1 \\ 0 & 0 & x & 5 \\ 0 & 2 & 3 & 0 \end{bmatrix}.$$

Find the value of x such that the dimension of the left nullspace of A is greater than 0 .

Your Answer: | Correct Answer: -20 | Not Attempted | Time taken: 00min 54sec | Discuss

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